

PRACTICE TEST FOR TEST 3 (STAT-461/561)

Complete Solutions

(1) Identify each of the following statements as TRUE or FALSE:

- (a) Two random variables X and Y are independent if and only if their joint p.d.f. is the product of their individual (or marginal) p.d.f.s;
- (b) For a discrete random vector (X, Y) where each of X and Y takes two values (x_1 and x_2 for X , y_1 and y_2 for Y) with

$$P(X = x_1, Y = y_1) = p_{11}, \quad P(X = x_1, Y = y_2) = p_{12},$$

$$P(X = x_2, Y = y_1) = p_{21}, \quad P(X = x_2, Y = y_2) = p_{22},$$

the random variables X and Y are independent if $p_{11}p_{22} - p_{12}p_{21} = 0$;

- (c) If X and Y are two random variables, each having mean μ and variance σ^2 , the two random variables $X + Y$ and $X - Y$ are uncorrelated;
- (d) If the correlation between two random variables X and Y is r , the correlation between $U = aX + b$ and $V = cY + d$ is $(acr + bd)$.

Solution.

- (a) **True.** For continuous random variables, independence is equivalent to

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

for all (x, y) in the support.

- (b) **True.** In the 2×2 discrete case, independence is equivalent to

$$p_{11}p_{22} = p_{12}p_{21},$$

i.e.

$$p_{11}p_{22} - p_{12}p_{21} = 0.$$

- (c) **True.** Since

$$\text{Cov}(X + Y, X - Y) = \text{Cov}(X, X) - \text{Cov}(X, Y) + \text{Cov}(Y, X) - \text{Cov}(Y, Y),$$

we get

$$\text{Cov}(X + Y, X - Y) = \text{Var}(X) - \text{Var}(Y) = \sigma^2 - \sigma^2 = 0.$$

So $X + Y$ and $X - Y$ are uncorrelated.

- (d) **False.** For $a \neq 0$ and $c \neq 0$,

$$\text{Corr}(aX + b, cY + d) = \frac{ac}{|ac|} \text{Corr}(X, Y) = \text{sign}(ac) r,$$

not $acr + bd$.

- (2) If a continuous random variable X has CDF $F(x) = x^4$ for $x \in [0, 1]$ and $= 0$ for $x < 0$ and $= 1$ for $x > 1$, find the corresponding pdf. Can you identify it as belonging to a known family of densities? Find the mean and the variance of this X . Also, find its median.

Solution. For $0 \leq x \leq 1$,

$$F(x) = x^4.$$

Therefore the pdf is

$$f(x) = F'(x) = 4x^3, \quad 0 \leq x \leq 1,$$

and $f(x) = 0$ otherwise.

This is a Beta density with parameters $\alpha = 4$ and $\beta = 1$.

Hence

$$E(X) = \frac{\alpha}{\alpha + \beta} = \frac{4}{5} = 0.8,$$

and

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{(4)(1)}{5^2 \cdot 6} = \frac{4}{150} = \frac{2}{75}.$$

The median m satisfies $F(m) = 0.5$, so

$$m^4 = 0.5.$$

Thus

$$m = 0.5^{1/4} \approx 0.8409.$$

- (3) If $X_1, \dots, X_n$ are i.i.d. Gamma(3, 5) random variables, find the mean and the variance of

$$X = \sum_{i=1}^n (c_i X_i + b_i)$$

for some constants $c_1, \dots, c_n$ and $b_1, \dots, b_n$.

Solution. Using the Gamma(α, β) parameterization with shape $\alpha = 3$ and scale $\beta = 5$,

$$E(X_i) = \alpha\beta = 3(5) = 15, \quad \text{Var}(X_i) = \alpha\beta^2 = 3(5^2) = 75.$$

Therefore

$$E(c_i X_i + b_i) = c_i E(X_i) + b_i = 15c_i + b_i,$$

and

$$\text{Var}(c_i X_i + b_i) = c_i^2 \text{Var}(X_i) = 75c_i^2.$$

Hence

$$E\left(\sum_{i=1}^n (c_i X_i + b_i)\right) = \sum_{i=1}^n (15c_i + b_i),$$

and, since the X_i are independent,

$$\text{Var}\left(\sum_{i=1}^n (c_i X_i + b_i)\right) = \sum_{i=1}^n 75c_i^2.$$

(4) If (X, Y) is a random vector having the joint pdf

$$f(x, y) = \frac{1}{64}xy^2e^{-(x+y)/2}$$

for $x > 0$ and $y > 0$, derive the marginal pdf of X and that of Y . What are the marginal mean and variance for each of X and Y ? What is the correlation between X and Y ? What is $E(X^2Y^2)$? [**Hint:** In order to avoid lengthy answers to the last two questions, check if X and Y are independent.]

Solution. First rewrite the joint pdf as

$$f(x, y) = \left(\frac{1}{4}xe^{-x/2}\right)\left(\frac{1}{16}y^2e^{-y/2}\right), \quad x > 0, y > 0.$$

So $f(x, y) = f_X(x)f_Y(y)$, which shows that X and Y are independent.

The marginal pdf of X is

$$f_X(x) = \int_0^\infty \frac{1}{64}xy^2e^{-(x+y)/2} dy = \frac{1}{4}xe^{-x/2}, \quad x > 0.$$

Thus $X \sim \Gamma(2, 2)$, so

$$E(X) = 2(2) = 4, \quad \text{Var}(X) = 2(2^2) = 8.$$

The marginal pdf of Y is

$$f_Y(y) = \int_0^\infty \frac{1}{64}xy^2e^{-(x+y)/2} dx = \frac{1}{16}y^2e^{-y/2}, \quad y > 0.$$

Thus $Y \sim \Gamma(3, 2)$, so

$$E(Y) = 3(2) = 6, \quad \text{Var}(Y) = 3(2^2) = 12.$$

Since X and Y are independent,

$$\text{Corr}(X, Y) = 0.$$

Also,

$$E(X^2Y^2) = E(X^2)E(Y^2).$$

Now

$$E(X^2) = \text{Var}(X) + [E(X)]^2 = 8 + 4^2 = 24,$$

and

$$E(Y^2) = \text{Var}(Y) + [E(Y)]^2 = 12 + 6^2 = 48.$$

Therefore

$$E(X^2Y^2) = 24 \cdot 48 = 1152.$$

- (5) If the random vector (X, Y) has the joint pdf $f(x, y) = k(1 - x)$ on $0 \leq y \leq x \leq 1$ and $= 0$ elsewhere, what must k be for $f(x, y)$ to be a legitimate joint density? Find the marginal densities of X and Y . Find the conditional density function of X , given that $Y = y$. Are X and Y independent?

Solution. To make $f(x, y)$ a joint density, we need

$$\int_0^1 \int_0^x k(1 - x) dy dx = 1.$$

So

$$k \int_0^1 x(1 - x) dx = k \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = k \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{k}{6} = 1.$$

Hence

$$k = 6.$$

Therefore the joint density is

$$f(x, y) = 6(1 - x), \quad 0 \leq y \leq x \leq 1.$$

The marginal density of X is

$$f_X(x) = \int_0^x 6(1 - x) dy = 6x(1 - x), \quad 0 \leq x \leq 1.$$

The marginal density of Y is

$$f_Y(y) = \int_y^1 6(1 - x) dx = 6 \left[x - \frac{x^2}{2} \right]_y^1 = 3 - 6y + 3y^2 = 3(1 - y)^2, \quad 0 \leq y \leq 1.$$

For $0 \leq y \leq 1$, the conditional density of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{6(1 - x)}{3(1 - y)^2} = \frac{2(1 - x)}{(1 - y)^2}, \quad y \leq x \leq 1.$$

Since the support is triangular, not rectangular, X and Y cannot be independent. Equivalently,

$$f(x, y) \neq f_X(x)f_Y(y).$$

So X and Y are **not** independent.

- (6) If the joint p.m.f. of the random vector (X, Y) is given by the following table, find the correlation between X and Y . Are X and Y independent? If not, write down the conditional p.m.f. table of X given that $Y = 4$.

	$Y = 2$	$Y = 4$
$X = 0$	0.15	0.25
$X = 5$	0.40	0.10
$X = 10$	0.03	0.07

Solution. The marginal pmf of X is

$$P(X = 0) = 0.40, \quad P(X = 5) = 0.50, \quad P(X = 10) = 0.10.$$

So

$$E(X) = 0(0.40) + 5(0.50) + 10(0.10) = 3.5,$$

and

$$E(X^2) = 0^2(0.40) + 5^2(0.50) + 10^2(0.10) = 22.5.$$

Hence

$$\text{Var}(X) = 22.5 - (3.5)^2 = 10.25.$$

The marginal pmf of Y is

$$P(Y = 2) = 0.58, \quad P(Y = 4) = 0.42.$$

So

$$E(Y) = 2(0.58) + 4(0.42) = 2.84,$$

and

$$E(Y^2) = 2^2(0.58) + 4^2(0.42) = 9.04.$$

Hence

$$\text{Var}(Y) = 9.04 - (2.84)^2 = 0.9744.$$

Also,

$$E(XY) = 5 \cdot 2 \cdot 0.40 + 5 \cdot 4 \cdot 0.10 + 10 \cdot 2 \cdot 0.03 + 10 \cdot 4 \cdot 0.07 = 9.4.$$

Thus

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 9.4 - (3.5)(2.84) = -0.54.$$

Therefore

$$\text{Corr}(X, Y) = \frac{-0.54}{\sqrt{(10.25)(0.9744)}} \approx -0.171.$$

X and Y are not independent; for example,

$$P(X = 0, Y = 2) = 0.15 \neq P(X = 0)P(Y = 2) = 0.40(0.58) = 0.232.$$

Since

$$P(Y = 4) = 0.25 + 0.10 + 0.07 = 0.42,$$

the conditional pmf values are

$$P(X = 0 \mid Y = 4) = \frac{0.25}{0.42} = \frac{25}{42},$$

$$P(X = 5 \mid Y = 4) = \frac{0.10}{0.42} = \frac{5}{21},$$

and

$$P(X = 10 \mid Y = 4) = \frac{0.07}{0.42} = \frac{1}{6}.$$

So the conditional pmf table is

x	$P(X = x \mid Y = 4)$
0	$\frac{25}{42}$
5	$\frac{5}{21}$
10	$\frac{1}{6}$

- (7) Let $\{Y_1, \dots, Y_{51}\}$ be a simple random sample from a Normal population with $\mu = 150$ and $\sigma = 5$. What's the pdf of $\bar{Y}$? What's the pdf of $2s^2$? [s^2 = sample variance]

Solution. Since the sample comes from a Normal(150, 5²) population,

$$\bar{Y} \sim \text{Normal}\left(150, \frac{5^2}{51}\right) = \text{Normal}\left(150, \frac{25}{51}\right).$$

Therefore the pdf of $\bar{Y}$ is

$$f_{\bar{Y}}(y) = \frac{1}{\sqrt{2\pi(25/51)}} \exp\left(-\frac{(y-150)^2}{2(25/51)}\right), \quad -\infty < y < \infty.$$

Equivalently,

$$f_{\bar{Y}}(y) = \frac{\sqrt{51}}{5\sqrt{2\pi}} \exp\left(-\frac{51(y-150)^2}{50}\right), \quad -\infty < y < \infty.$$

Also, for a Normal sample,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Here $n = 51$ and $\sigma^2 = 25$, so

$$\frac{50S^2}{25} = 2S^2 \sim \chi_{50}^2.$$

Hence the pdf of $2S^2$ is the chi-square pdf with 50 degrees of freedom:

$$f_{2S^2}(t) = \frac{1}{2^{25}\Gamma(25)} t^{24} e^{-t/2}, \quad t > 0.$$

- (8) The daily total milk output from a small Wisconsin dairy farm is Normally distributed with mean 84 gallons and standard deviation 3.6 gallons. What are the chances that on a randomly selected day, the milk output will (a) exceed 90 gallons? (b) be at most 75 gallons? (c) be between 80 and 85 gallons?

Solution. Let X be the daily total milk output. Then

$$X \sim \text{Normal}(84, 3.6^2).$$

(a)

$$P(X > 90) = P\left(Z > \frac{90 - 84}{3.6}\right) = P(Z > 1.67) = 1 - 0.9525 = 0.0475.$$

(b)

$$P(X \leq 75) = P\left(Z \leq \frac{75 - 84}{3.6}\right) = P(Z \leq -2.50) = 0.0062.$$

(c)

$$P(80 < X < 85) = P\left(\frac{80 - 84}{3.6} < Z < \frac{85 - 84}{3.6}\right) = P(-1.11 < Z < 0.28).$$

Using the Normal table,

$$P(-1.11 < Z < 0.28) = 0.6103 - 0.1335 = 0.4768.$$

- (9) In #8 above, what is the 90th percentile of daily total milk output? On 80% of the days, the total milk output is at least how much?

Solution. For the 90th percentile, use $z_{0.90} \approx 1.28$. Then

$$\frac{x - 84}{3.6} = 1.28$$

gives

$$x = 84 + 1.28(3.6) = 88.608.$$

So the 90th percentile is about

88.608 gallons.

For the second part, we need the 20th percentile because

$$P(X \geq x) = 0.80 \iff P(X \leq x) = 0.20.$$

Using $z_{0.20} \approx -0.84$,

$$\frac{x - 84}{3.6} = -0.84,$$

so

$$x = 84 - 0.84(3.6) = 80.976.$$

Thus on 80% of days, the output is at least about

80.976 gallons.

- (10) In #8 above, if you took a random sample of $n = 81$ days and computed the sample average $\bar{X}$ of milk output, what are the chances that $\bar{X}$ would be (a) above 85 gallons? (b) below 82.8 gallons? (c) between 83 and 85 gallons?

Solution. Since

$$\bar{X} \sim \text{Normal}\left(84, \frac{3.6^2}{81}\right) = \text{Normal}(84, 0.16),$$

we have

$$\text{SD}(\bar{X}) = \frac{3.6}{9} = 0.4.$$

(a)

$$P(\bar{X} > 85) = P\left(Z > \frac{85 - 84}{0.4}\right) = P(Z > 2.5) = 1 - 0.9938 = 0.0062.$$

(b)

$$P(\bar{X} < 82.8) = P\left(Z < \frac{82.8 - 84}{0.4}\right) = P(Z < -3) = 0.0013.$$

(c)

$$P(83 < \bar{X} < 85) = P\left(\frac{83 - 84}{0.4} < Z < \frac{85 - 84}{0.4}\right) = P(-2.5 < Z < 2.5).$$

Therefore

$$P(-2.5 < Z < 2.5) = 0.9938 - 0.0062 = 0.9876.$$

- (11) If X and Y are independent random variables with means 23 and 17, and standard deviations 2 and 4 respectively, find the means and variances of (a) $15X$, (b) $20X - 12Y$, (c) $5X + 8Y$.

Solution. Since

$$E(X) = 23, \quad E(Y) = 17, \quad \text{Var}(X) = 2^2 = 4, \quad \text{Var}(Y) = 4^2 = 16,$$

and X, Y are independent, we obtain:

(a)

$$E(15X) = 15E(X) = 345, \quad \text{Var}(15X) = 15^2 \text{Var}(X) = 15^2(4) = 900.$$

(b)

$$E(20X - 12Y) = 20E(X) - 12E(Y) = 20(23) - 12(17) = 256,$$

and

$$\text{Var}(20X - 12Y) = 20^2 \text{Var}(X) + (-12)^2 \text{Var}(Y) = 400(4) + 144(16) = 3904.$$

(c)

$$E(5X + 8Y) = 5E(X) + 8E(Y) = 5(23) + 8(17) = 251,$$

and

$$\text{Var}(5X + 8Y) = 5^2 \text{Var}(X) + 8^2 \text{Var}(Y) = 25(4) + 64(16) = 1124.$$

- (12) In 200 tosses of a coin with $P(H) = \frac{3}{4}$, what are the chances of getting fewer than 165 heads?

Solution. Let X be the number of heads. Then

$$X \sim \text{Binomial}(200, 0.75).$$

The exact probability is

$$P(X < 165) = \sum_{k=0}^{164} \binom{200}{k} (0.75)^k (0.25)^{200-k} \approx 0.9927.$$

Using the Normal approximation employed in the provided work,

$$\mu = np = 150, \quad \sigma^2 = np(1 - p) = 37.5.$$

Then

$$P(X < 165) = P(X \leq 164) \approx P\left(Z \leq \frac{164.5 - 150}{\sqrt{37.5}}\right) = P(Z \leq 2.37) = 0.9911.$$

So the normal-approximation answer is

$$P(X < 165) \approx 0.9911,$$

while the exact binomial probability is about 0.9927.

- (13) For a simple random sample of size n from a population with a Poisson distribution, derive the maximum likelihood estimator of the population mean. What is the MOM estimator of the population mean?

Solution. Let $X_1, \dots, X_n$ be a simple random sample from a Poisson population with mean μ . For a Poisson random variable,

$$P(X = k) = \frac{e^{-\mu} \mu^k}{k!}, \quad k = 0, 1, 2, \dots$$

The likelihood is

$$L(\mu; x_1, \dots, x_n) = \prod_{i=1}^n \frac{e^{-\mu} \mu^{x_i}}{x_i!} = \frac{e^{-n\mu} \mu^{\sum_{i=1}^n x_i}}{x_1! \cdots x_n!}.$$

The log-likelihood is

$$\ell(\mu) = -n\mu + \left(\sum_{i=1}^n x_i\right) \ln \mu - \sum_{i=1}^n \ln(x_i!).$$

Differentiate and set equal to zero:

$$\frac{d\ell}{d\mu} = -n + \frac{\sum_{i=1}^n x_i}{\mu} = 0.$$

So

$$\hat{\mu}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}.$$

For the method of moments, since the population mean is μ , equating it to the sample mean gives

$$\hat{\mu}_{\text{MOM}} = \bar{X}.$$

Therefore

$$\boxed{\hat{\mu}_{\text{MLE}} = \bar{X}}, \quad \boxed{\hat{\mu}_{\text{MOM}} = \bar{X}}.$$

- (14) For a simple random sample of size n from a population having a $\text{Normal}(\mu, \sigma^2)$ density, find the maximum likelihood estimators of the population mean and the population variance. What are the MOM estimators of those two population parameters?

Solution. For a $\text{Normal}(\mu, \sigma^2)$ sample, the likelihood is

$$L(\mu, \sigma^2; x_1, \dots, x_n) = (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Differentiating with respect to μ and setting equal to zero gives

$$\hat{\mu}_{\text{MLE}} = \bar{X}.$$

Substituting $\mu = \bar{X}$ and maximizing with respect to σ^2 gives

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2.$$

For the method of moments, use the first two raw moments:

$$E(X) = \mu, \quad E(X^2) = \mu^2 + \sigma^2.$$

Equating sample moments to population moments gives

$$\hat{\mu}_{\text{MOM}} = \bar{X},$$

and

$$\hat{\sigma}_{\text{MOM}}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

So

$$\boxed{\hat{\mu}_{\text{MLE}} = \bar{X}}, \quad \boxed{\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2},$$

and

$$\boxed{\hat{\mu}_{\text{MOM}} = \bar{X}}, \quad \boxed{\hat{\sigma}_{\text{MOM}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}.$$

- (15) If a random sample of 400 people was asked the question “Do you think that using AI is a good idea for school students — yes or no?” and 251 of them said “no”, what is a point estimate of the (unknown) population proportion of people believing “no”? Construct a 98% confidence interval for this population proportion.

Solution. The point estimate is the sample proportion:

$$\hat{p} = \frac{251}{400} = 0.6275.$$

Using $z_{0.01} \approx 2.33$, an approximate 98% confidence interval is

$$\hat{p} \pm 2.33 \sqrt{\frac{\hat{p}(1 - \hat{p})}{400}}.$$

So

$$0.6275 \pm 2.33 \sqrt{\frac{0.6275(0.3725)}{400}} \approx 0.6275 \pm 0.0563.$$

Hence the interval is

$$\boxed{(0.5712, 0.6838)}.$$

- (16) If a 98% confidence interval for an unknown population proportion p is to be 0.2 units wide, find the sample size needed.

Solution. A total width of 0.2 means the margin of error is

$$E = 0.1.$$

Since p is unknown, use the conservative value $p(1 - p) \leq 0.25$. Then

$$E = z_{0.01} \sqrt{\frac{0.25}{n}}$$

with $z_{0.01} \approx 2.33$. So

$$0.1 = 2.33 \sqrt{\frac{0.25}{n}}$$

which gives

$$n = \frac{(2.33)^2(0.25)}{0.1^2} = \left(\frac{2.33}{0.2}\right)^2 = 135.72.$$

Round up to get

$$\boxed{n = 136}.$$

- (17) The hourly wages in a particular industry are normally distributed with mean \$13.20 and standard deviation \$2.50. A company in this industry employs 40 workers, paying them an average of \$12.20 per hour. At a 1% significance level, can this company be accused of paying substandard wages? What is the p-value of this test?

Solution. Let μ be the company's true mean hourly wage. We test

$$H_0 : \mu \geq 13.20 \quad \text{vs.} \quad H_1 : \mu < 13.20.$$

Because the population standard deviation is known, use the z statistic:

$$z = \frac{\bar{x} - 13.2}{2.5/\sqrt{40}} = \frac{12.2 - 13.2}{2.5/\sqrt{40}} = -2.529.$$

At the 1% significance level for a left-tailed test, the critical value is

$$z_{0.01} = -2.33.$$

Since

$$-2.529 < -2.33,$$

we reject H_0 .

The p-value is

$$P(Z \leq -2.529) \approx 0.0057.$$

Therefore there is sufficient evidence at the 1% level to accuse the company of paying substandard wages.

- (18) A simple random sample of 25 students was taken from among all the students enrolled in the University of Akron's remedial mathematics course (for freshmen only), which showed a sample average high-school GPA of 2.79 and a sample standard deviation of 0.12. What is a 95% confidence interval for the true, unknown population mean high-school GPA of all freshmen enrolled in that course?

Solution. Here $n = 25$, $\bar{x} = 2.79$, and $s = 0.12$. Since σ is unknown and the population is assumed Normal, use a t interval:

$$\bar{x} \pm t_{0.025, 24} \frac{s}{\sqrt{n}}.$$

Using $t_{0.025, 24} = 2.064$,

$$2.79 \pm 2.064 \left(\frac{0.12}{5} \right) = 2.79 \pm 0.0495.$$

Thus the 95% confidence interval is

$$\boxed{(2.7405, 2.8395)}.$$

- (19) In a simple random sample of 50 large and mid-sized retailers from the greater Cleveland area, 29 believe that the upcoming holiday season will be better than last year in terms of sales volume and revenue. At a 1% significance level, test whether a majority of all large and mid-sized retailers in that area believe so. What is the p-value of this test?

Solution. Let p be the true proportion of such retailers who believe the upcoming holiday season will be better. Since “majority” means more than 0.5, test

$$H_0 : p \leq 0.5 \quad \text{vs.} \quad H_1 : p > 0.5.$$

The sample proportion is

$$\hat{p} = \frac{29}{50} = 0.58.$$

Using the normal approximation under $H_0 : p = 0.5$,

$$z = \frac{\hat{p} - 0.5}{\sqrt{0.5(1 - 0.5)/50}} = \frac{0.08}{\sqrt{0.25/50}} \approx 1.13.$$

At the 1% significance level for a right-tailed test, the critical value is

$$z_{0.99} = 2.33.$$

Since

$$1.13 < 2.33,$$

we do not reject H_0 .

The p-value is

$$P(Z \geq 1.13) \approx 0.1292.$$

There is not enough evidence at the 1% level to conclude that a majority of these retailers believe the upcoming holiday season will be better.